13/02/2026 Tutorial: 5

SVM classifier

Input Features:

1) Number of suspicious words.
2) Number of External Links
3) Email Length (words)
4) Attachment Present (0/1)

Target Variable:

. class (Spam = +1, Not Spam = -1)

Suspicious words (x_1)	Length (x_2)	Class
6	100	+1
1	350	-1
4	150	+1
0	420	-1
5	120	+1
2	260	-1

✓ $x_1 \rightarrow (0, 2) \rightarrow$ Emails — Non-Spam (-1)

✓ $x_2 \rightarrow (>2) \rightarrow$ Emails — Spam (+1)

$x_2 \rightarrow (100 - 150) \rightarrow +1$

$x_2 \rightarrow (260 - 420) \rightarrow -1$

25/2/2026

** SVM

(x_1) Suspicious words	(x_2) Links	(x_3) Attachment ~~length~~	class
6	3	1	-1
1	0	0	-1
4	2	1	+1
0	0	0	-1
5	2	1	+1
(w_1)	(w_2)	(w_3)	

⇒ $\boxed{\begin{array}{l} -1 \text{ non spam} \\ 1 \text{ spam} \end{array}}$ and bias(b)

⇒ Predict y(class): weighted sum

① $f(x) = z = \underset{(b)}{w_0} + w_1x_1 + w_2x_2 + w_3x_3$

(or)

$f(x) = \underbrace{w^T x}_{(w_1, w_2, w_3)} + b$ (or) $w^T x + w_0$

$$w = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

② check quantity

$$y_i (w_0 + w_1x_1 + w_2x_2 + w_3x_3) \geq 1$$

(perfect classification)

NOTE: $y_i [f(x)] \geq 1 \rightarrow$ P.C

$y_i [f(x)] < 1 \rightarrow$ classification fails.

Suppose:

$f(x)_1 =$

y_i [f

-1

$\mathcal{L}(w, b$

↳ cost f

cos

if class

the we

w_i ne

$\frac{\partial L}{\partial w}$

Imp w_i ne

w_1 new

w_1 ne

w_2 new

$$f(x)_1 = x_1 w_1 + x_2 w_2 + x_3 w_3 + w_0$$
$$= (6)(1) + 3(1) + (1)(1) + 0$$
$$= 10.$$

$$y_i \underbrace{[f(x)]}_{10} = -1 \times 10 = -10 < 1$$

-1

classification fails

$$\mathcal{L}(w, b, c) = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{m} c_i \left[y_i (w_i x + b) - 1\right]$$

c_i → classification error

↳ cost function (or) loss function.

if classification fails we have to update the weights:

$$w_i^{new} = w_i^{old} - \eta \times \frac{\partial L}{\partial w}$$

$$\frac{\partial L}{\partial w} = w - \sum_{i=1}^{n} c_i y_i x_{ij}$$

$$\boxed{w_i^{new} = w_i^{old} - \eta \times \left[w_i^{old} - \sum_i c_i y_i x\right]}$$

Imp

$\boxed{c = 1}$ $y_i = -1, x =$

$$w_1^{new} = 1 - 1 \times \left[1 - 1((-1)(6))\right]$$

$$\boxed{w_1^{new} = -6}$$

$$w_2^{new} = 4 - 1 \times \left[4 - 1(-1)(3)\right]$$

$$= -3$$

$$w_3^{new} = w_3^{old} - \eta \left[w_3^{old} - \sum a_i y_i x_i \right]$$

$$= 0 - 1 \left[1 - \left[(1)(-1)(1) \right] \right]$$

$$= -1$$

w_1^{new} ; w_2^{new} w_3^{new}

$\downarrow$ $\downarrow$ $\downarrow$

-6 -3 -1

NOTE

$\boxed{w_0 = b}$

updating bias.

$$w_0^{new} = w_0^{old} - \eta \times \frac{\partial L}{\partial w_0}$$

$$\boxed{\frac{\partial L}{\partial w_0} = \frac{\partial L}{\partial b} = -\sum_{i=1}^{N} a_i y_i}$$

$$w_0^{new} = w_0^{old} - \eta * \left[-\sum_{i=1}^{N} a_i y_i \right]$$

Imp $$\boxed{w_0^{new} = w_0^{old} + \eta \sum_{i=1}^{N} a_i y_i}$$

$$w_0^{new} = 0 + 1 (1)(-1)$$

$$w_0^{new} = -1$$

$w_0 = -1$, $w_1 = -6$, $w_2 = -3$, $w_3 = 1$

↳ after 1st set of inputs, updated weights and bias.

$x_1 = 1 \quad x_2 = 0 \quad x_3 = 0$

(D) $f(x) = x_1 w_1 + x_2 w_2 + x_3 w_3 + w_0$

$= (1)(-6) + (0)(-3) + (0)(1) - 1$

$= -6 - 1 = -7$

$y_i[f(x)] = -1 \times -7 = 7 \geq 1 = 1$

perfect classification

loss function: $L = \frac{1}{2}\|w\|^2$

$w_i^{new} = w_i^{old} - \eta \frac{\partial L}{\partial w}$

↳ our prediction is perfect so, $c = 0$

$\boxed{\eta = 0.1}$

$\boxed{w_i^{new} = w_i^{old} - \eta w_i^{old}}$

$= -6 - 1(-6) = 0$

$w_2^{new} = -3 - (0.1)(-3) = 0$

$w_3^{new} = 0$

No updation of bias because we will not take $\frac{\partial L}{\partial w_0}$ in loss function.

$\boxed{w_0^{new} = w_0^{old}}$

(3) $w_0 = -1 \quad w_1 = 0 \quad w_2 = 0 \quad w_3 = 0$

$$z = x_1 w_1 + x_2 w_2 + x_3 w_3 + w_0$$
$$= -1$$

$$f(x) = z = -1$$

$$y_i [f(x)] = 1 \times -1 = -1 < 1$$

classification fails.

→ <u>margin will be very high.</u>

if learning rate is 1 the weights will be zero so, for calculating margin at the end it will get ∞ so, It will ~~take~~ be high

<u>margin</u> should (goal) increase to have good classification values

calculate margin after every update of weights.

$$\text{Margin} = \frac{2}{\|w\|}$$

$$\boxed{\text{Margin} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2}}}$$